CONFIDENTIA
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SCHOOL OF COMPUTING
Faculty of Engineering

UNIVERSITI TEKNOLOGI MALAYSIA

MID-TERM TEST SEMESTER I 2019 / 2020

SUBJECT CODE	**:**	**SCSD2523**
SUBJECT NAME	**:**	**DATABASE**
SECTION	**:**	**1, 2 (SCSJ/SCSV/SCSB/SCSR)**
TIME	**:**	**8:00 P.M. – 10:30 P.M. (2 HOURS 30 MINUTES)**
DATE/DAY	**:**	**05 NOVEMBER 2019**
VENUES	**:**	**LECTURE HALL (DK4-6), P19**

INSTRUCTIONS:

1. This test book consists of **THREE** (3) sections:
   a. Section A: True / False questions. (10 Marks)
   b. Section B: Structured questions. (60 Marks)
   c. Section C: Case Study. (30 Marks)
2. Answer **ALL** questions in the answer booklet provided.

(Please Write Your Lecturer Name and Section in Your Answer Booklet)

Name	
I/C No.	
Year / Course	
Section	
Lecturer Name	

This question paper consists of **NINE (9)** printed pages excluding this page.

SECTION A: TRUE / FALSE QUESTIONS **10 MARKS**

INSTRUCTION: Answer **ALL** questions.

Please write your answer (**T/F**) in the answer booklet provided.

1. Database application enables users to define, create and maintain the database and provide controlled access to the database. TRUE / FALSE

2. In relational model, a row is usually referred as a "tuple". TRUE / FALSE

3. The ANSI-SPARC Three-Level architecture consists of Conceptual level, Internal level and Physical level. TRUE / FALSE

4. Cardinality is referred as the number of tuples in a relation. TRUE / FALSE

5. A table can have multiple primary keys. TRUE / FALSE

6. In the database planning stage, boundaries of the database application are described and defined. TRUE / FALSE

7. SQL commands is used to access the database. TRUE / FALSE

8. In Enhanced Entity Relationship Modeling (EERM), the relationship between superclass and subclass is one-to-many. TRUE / FALSE

9. In EERM, disjoint constraint (OR) means that an entity occurrence can be a member for more than one subclass. TRUE / FALSE

10. Data Manipulation Language (DML) is used to create, read, update and delete records in the database and to navigate between different records and types of records. TRUE / FALSE

SECTION B: STRUCTURED QUESTIONS **60 MARKS**

INSTRUCTION: Answer **ALL** questions.

Please write your answer in the answer booklet provided.

QUESTION 1: (10 MARKS)

a) **Figures 1.1** and **1.2** represents different approaches in managing data.

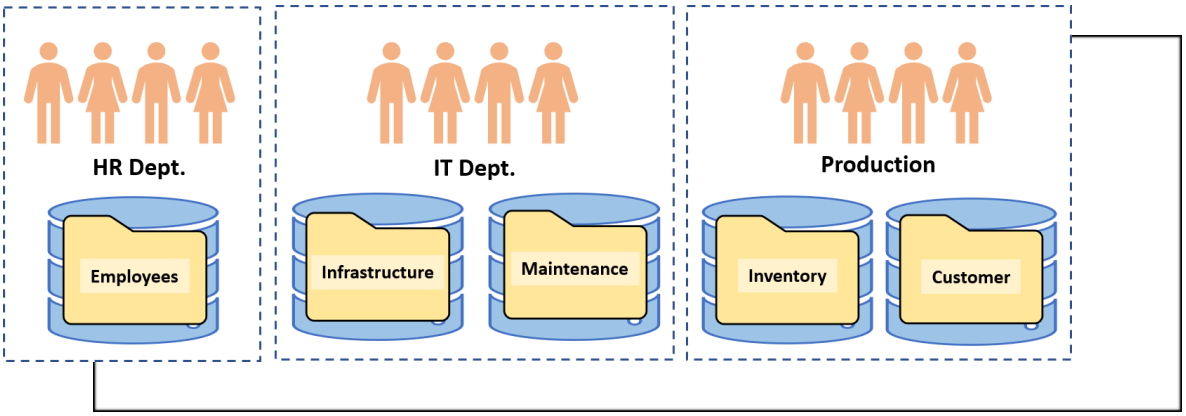


Figure 1.1

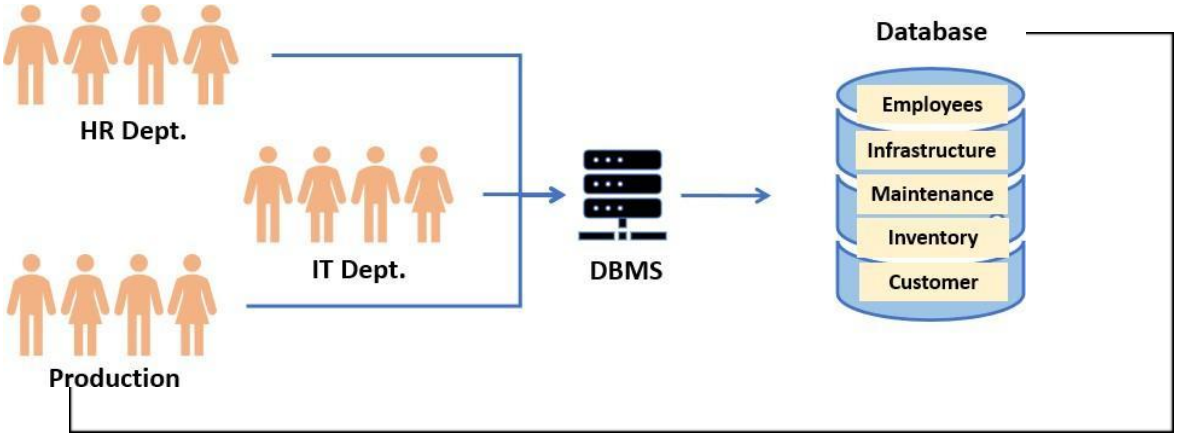


Figure 1.2

From the figures above, answer the following questions.

i) State the approaches implemented in **Figure 1.1** and **Figure 1.2**. (2 Marks)

ii) Describe the approaches mentioned in (i). (4 Marks)

iii) State **TWO** (2) limitations that existed in **EACH** of the approaches mentioned in (i).

(4 Marks)

QUESTION 2: (10 MARKS)

a) **Tables 2.1** and **2.2** below shows the Employee and Inventory relation, which has *empID* and *inventID* as the primary key for the each of the relation respectively.

Table 2.1: Employee

empID	empName	position	inventID
FSA104	Royal	Supervisor	INVA882
FTB448	Albert	Cleaner	INVC344
FGA661	Hall	Guard	INVT446

Table 2.2: Inventory

inventID	invName	invType	dateReceived
INVC344	Kettle	Electrical	30 / 04 / 2019
INVT446	Pen	Stationary	06 / 05 / 2019
INVA882	Shirt	Attire	01 / 02 / 2019

Answer the following questions based on the given tables.

i) Write the relational schema for each of the relation in **Table 2.1** and **2.2**. (3 Marks)

ii) Do **Tables 2.1** and **2.2** enforce *entity integrity* and *referential integrity*? Justify your answer with examples from the tables. (4 Marks)

iii) Justify why *empID* can be chosen as Primary Key in Employee relation. (1 Mark)

iv) Views is a virtual relation that does not necessarily actually exist in the database but is produced upon request, at time of request. State **TWO** (2) purposes of views in Database Management System (DBMS). (2 Marks)

QUESTION 3: (10 MARKS)

a) Briefly explain about the **THREE** (3) database design phases. (3 Marks)

b) Explain the following stages in Database System Development Lifecycle (DBSDLC):

i) Database planning

ii) System definition

iii) Requirement collection and analysis

(3 Marks)

c) Explain about *centralized approach* and *view integration approach* in requirement collections and analysis stage. (4 Marks)

QUESTION 4: (15 MARKS)

a) **Figure 4.1** shows the relationships R1-R4 in a system of a convenience store. **Construct** separate ERD **FOR EACH** of the relationships R1-R4. Use proper UML notations and write all the multiplicities. (8 marks)

R1: A customer might purchase many products; A product might be purchased by many customers.
R2: A customer can have many memberships; Each membership belongs to only one customer.
R3: A customer might make many orders; Each order is made by only one customer.
R4: A product can exist in many orders; Each order consists of at least one product.

Figure 4.1

b) Explain the **participation** for every entity in **EACH** of the relationships b(i)-b(iii) shown in **Figure 4.2.** (3 Marks)

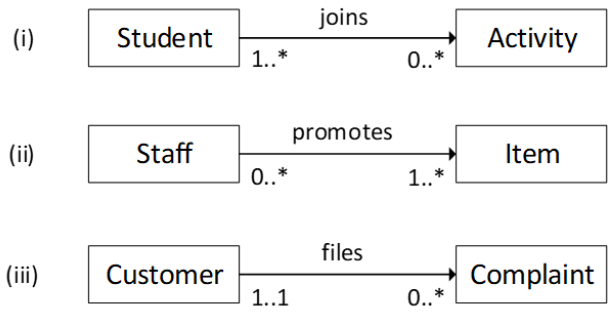


Figure 4.2

c) Based on the situation given below, fill the constraints c(i) and c(ii) for the Enhanced ERD (EERD) in **Figure 4.3**. (4 Marks)

" Mr. Cobby owns a service centre for repairing computer. His staffs are categorized as clerks, technicians and engineers, where each personnel have a distinct job scope. Clerks focus on clerical tasks; technician focus on the repair works while engineers look deeper into the problems encountered and feedback to the manufacturer for future improvements. In general, majority of the technicians can be divided into two groups, one focus on laptop products, one focus on desktop products. A technician currently only focusses one type of product at a time."

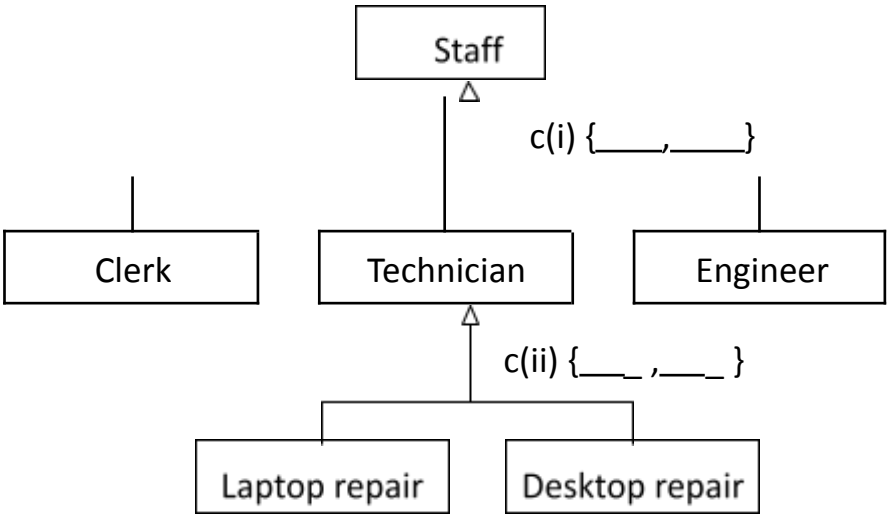

Staff
c(i) {_____,_____}
Clerk
Technician
Engineer
c(ii) {___ ,___ }
Laptop repair
Desktop repair

Figure 4.3

QUESTION 5: (15 MARKS)

Figures 5.1 shows the tables from part of a database held in a relational DBMS:

Hotel	(**hotelNo**, hotelName, city)
Room	(**roomNo**, **hotelNo**, type, price)
Booking	(**hotelNo**, **guestNo**, **dateFrom**, dateTo, roomNo)
Guest	(guestNo, guestName, guestAddress)

Figure 5.1

With specifications where:

- **Hotel** contains hotel details and **hotelNo** is the primary key.
- **Room** contains room details for each hotel and (**roomNo, hotelNo**) forms the primary key.
- **Booking** contains details of bookings and (**hotelNo, guestNo, dateFrom**) are the primary key.
- **Guest** contains guest details and **guestNo** is the primary key.

Based on the information in **Figure 5.1**, write the SQL statements for the following tasks:

a) Create the **Hotel** table with integrity enhancement features. (3 Marks)

b) Create the **Booking** table with integrity enhancement features. (3 Marks)

c) A new hotel named Impiana is opened at Kuala Lumpur with the hotel id of 25. Insert this new hotel's information into the **Hotel** table. (2 Marks)

d) Add a new attribute named "contact" into **Guest** table that only accept integer as input. (2 Marks)

e) List all the details for all rooms. (2 Marks)

f) List all the details of all hotels in Pulau Pinang. (3 Marks)

SECTION C: CASE STUDY **30 MARKS**

INSTRUCTIONS:

- Read the following case study carefully, then answer **ALL** questions (a) and (b).
- Please write your answer in the answer booklet provided.

As a junior database designer, your project manager requested you to construct a conceptual database design for a doctor-patient management system intended to be implemented in XYZ hospital. The aim is to design a database for the hospital to maintain the records of doctors, patients, prescriptions, and medications. The requirements are as follow:

- Every doctor's information such as doctorID, name, contact number, specialization and qualification are held by the hospital. The doctorID is unique for each doctor.
- All doctors in the hospital can be further categorized as *regular* and *on-call*. Regular doctors are those who are hired permanently by the hospital. On-call doctors are those who are invited from other hospitals when extra doctors are required in the hospital. Some regular doctors also invited as on-call doctors if required. Regular doctors will have the salary information while on-call doctors will have the hour-rate information.
- The hospital keeps track of each patient via a unique PatientID, together with their information such as name, contact number, address, gender, age and admission status.
- All patient admitted to the hospital can be divided into *local* and *international*, where local patients will have their identification number and international patients will have the additional information of country and passport number.
- The hospital also keeps track of the medicine via the medicineID, medicine name, supplier, and stock.

- The hospital also records the prescriptions gave by doctors, which consists of a unique prescriptionID, date of prescription, doctor's comment and medicine prescribed.
- In XYZ hospital, each patient is treated by at least one doctor at a time and each doctor will have at least one patient to take care of.
- Each doctor can give many prescriptions and each prescription is given by only one doctor specifically to one patient. A patient could have multiple prescriptions throughout the visits to the hospital.
- A prescription contains at least one type of medicine while each medicine can be prescribed by in multiple prescriptions.

a) Based on the case study, identify the **ENTITY**, **ATTRIBUTE** for each entity, and the corresponding **PRIMARY KEY** (if any). List your answer in a table form as following example:

Entity	**Attribute**
entityName	attribName, attrib2Name

(10 Marks)

b) Construct a Conceptual Entity Relationship Diagram (ERD) based on the case study in proper UML notation. State your assumptions if any.

(20 Marks)